

Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity

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Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here, we present a modelled, quantum-guided agrivoltaic system incorporating a semi-transparent organic photovoltaic layer coupled with a plasmonic nanoparticle-on-mirror (NPoM) cavity that acts as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. Secure IoT telemetry is maintained over a BB84 quantum link. Finally, cooperative economic modeling for a Cameroonian micro-module demonstrates a payback period of 4.23 yr, identifying conditions under which the multifunctional net benefit stays positive for the modelled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics, quantum digital twin

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1 Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cul-
 2 tivation to mitigate the land-use competition between food production and
 3 renewable energy generation [1, 2]. Land-use optimisation and mobile-PV
 4 foundations for this field were established in *Applied Energy* [3, 4], and recent
 5 syntheses systematize overhead, vertical, and tracking typologies with land-
 6 equivalent-ratio and levelized-cost metrics [5]. Across these studies, three
 7 gaps recur and remain unresolved. (i) The optical architecture is *static*:
 8 semi-transparent PV claddings act as fixed spectral filters that attenuate ir-
 9 radiance uniformly, trading electrical output for crop biomass without opti-
 10 mization of the spectral light-sharing window [6, 7]; dynamic, physics-guided
 11 spectral management has not been transposed into an operating agrivoltaic
 12 stack. (ii) Techno-economic and lifecycle assessments remain decoupled from
 13 the underlying device physics: electricity, crop, water, and carbon accounts
 14 are evaluated subsystem by subsystem, so multifunctional trade-offs are de-
 15 scribed rather than co-optimized [8, 9]. (iii) The PV layer is treated purely
 16 as an energy harvester: in-situ crop-stress and contaminant sensing is bolted
 17 on as separate hardware rather than co-designed with the optical stack, de-
 18 spite the maturity of field-deployable SERS analysis [10], of quantum-enabled
 19 agricultural workflows [11], and of quantum-sensing readiness for precision
 20 agriculture [12]. The photosynthetic apparatus of green plants and bacteria
 21 has evolved under a spectrally structured photon flux, where specific wave-
 22 lengths drive charge separation while others dissipate as heat [13]. This spec-
 23 tral specificity allows the PV layer to transmit only the photosynthetically
 24 active wavelengths while absorbing the remainder, converting the food-energy
 25 conflict into a spectral co-design problem.

26 The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria
 27 serves as a model system for exploring this concept [14]. Its seven bac-
 28 teriochlorophyll *a* sites have been extensively characterized through two-
 29 dimensional electronic spectroscopy, revealing oscillatory coherence signa-
 30 tures that persist for hundreds of femtoseconds at cryogenic [15] and room
 31 temperature [16]. Although the functional relevance of these quantum beats
 32 remains debated, consensus has emerged that vibronic coupling — the hy-
 33 bridization of electronic transitions with underdamped protein vibrations —
 34 governs the flow of excitation energy [17, 18]. Non-Markovian frameworks
 35 such as the hierarchical equations of motion [19] and the process tensor hier-
 36 archy of pure states (PT-HOPS) [20, 21] capture these effects with demon-
 37 strated numerical convergence. Device-adjacent uses of exact open-quantum-
 38 system methods remain rare — the closest precedent is a general HEOM li-
 39 brary applied to light-harvesting and control problems without a photovoltaic
 40 product [22] — and photonic-quantum solar design and vibronic-screening

workflows likewise stop short of an operating device [23, 24].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [25]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. A related study from our group, published in the *Journal of Physical Chemistry Letters*, reports the filter, excitation-transport-rate, and single-pass-conversion-efficiency analysis [26]; the present work adds the monolithic photothermal-heating hardware, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [27], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [28]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [29] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and optomechanically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic quantum optimization onto a macroscopic multi-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman–Monteith model [30] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [31] for secure IoT telemetry. This multi-scale perspective [32] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modelling framework for quantum-guided agrivoltaics (Fig. 4).

The central question guiding this work is falsifiable: *can exact non-Markovian quantum dynamics (PT-HOPS) jointly optimize excitation-transport efficiency and power-conversion efficiency in a sensing-integrated*

79 *photothermal agrivoltaic stack — at what capital cost does the resulting*
80 *multifunctional net energy benefit (heat + power + crop + sensing, in $\$/W$*
81 *+ $\$/kg$ + $\$/m^3$) stay positive under field constraints, and which deployment*
82 *barriers are physical rather than economic?* Three falsifiable claim families
83 follow. **(F1) Microscopic:** dual-band spectral filtering at 750 nm and
84 820 nm prepares polaron-dressed states that extend exciton coherence by
85 50 % and raise the forward transfer yield by 25 % at 295 K. **(F2) Device:**
86 confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the
87 global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex
88 SERS diagnostics, despite the > 90 % local transport penalty quantified
89 across the mode-volume scan. **(F3) System:** the FAO-56 smart shield
90 cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and,
91 combined with OPV generation and cooperative economics, yields a Net
92 Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$ and a 4.23 yr payback at 30 %
93 subsidy for the modelled 500 m^2 cooperative case. We answer affirmatively
94 but conditionally: the bounds (single-exciton treatment, site-independent
95 baths, modelled fouling and matrix penalties, ± 10 % microclimate band)
96 are stated explicitly in the Limitations section.

97 Results

98 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm^{-1}).** Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	-87.7	30.8	-53.5	-70.7	81.1	39.7	12.0

99 We construct an 8-site model of the FMO complex, extending the stan-
100 dard 7-site framework [33] by including a coupling to the eighth bacteri-
101 ochlorophyll that bridges the baseplate to the reaction center (Site 8, ~ 200
102 cm^{-1} above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ is the irreversible reaction center trapping rate [34]. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. All 28 inter-site couplings are validated to be finite, and the Hamiltonian is verified to be non-Hermitian only on the trapping diagonal terms ([Section S1](#)).

NPoM plasmonic strong coupling

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) architecture [29], where a gold nanoparticle (diameter 80 nm) is separated from a gold mirror by a 1 nm graphene monolayer spacer. The graphene barrier prevents Dexter-type electron transfer while maintaining electromagnetic field enhancement. The NPoM cavity supports an intense localized surface plasmon mode that couples coherently to both the OPV exciton band and the FMO optical transitions.

The single-emitter coupling strength follows the cavity quantum electrodynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [29]. At the sub-nanometer mode volumes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1} to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the system in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plasmon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$ to prevent divergences. Recent advances in quantum-tunnel-junction plasmonic sources have demonstrated that such sub-nanometer cavities can be self-illuminating, eliminating the need for bulky external optics [35].

The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength g_0 . [Figure 1a](#) shows the energy level diagram.

132 *Non-Markovian quantum dynamics*

133 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
 134 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
 135 terms, time step $\Delta t = 0.2$ fs, and $n = 100$ disorder realizations ($\sigma = 50$ cm $^{-1}$
 136 static disorder) [20, 21]. The bath is modeled as a Drude–Lorentz over-
 137 damped component ($\lambda_D = 35$ cm $^{-1}$, $\gamma_D = 50$ cm $^{-1}$) plus 12 underdamped
 138 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
 139 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara
 140 truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$
 141 and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is
 142 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-
 143 aware scheduling for these high-order simulations are detailed and validated
 144 in our companion study [25].

145 Figure 1b shows the exciton population dynamics under broadband *vs.*
 146 dual-band filtered excitation. The filtered case (solid lines) maintains higher
 147 population on the trapping sites (Sites 3 and 4), leading to enhanced forward
 148 transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

149 where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

150 Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for
 151 broadband excitation — a relative enhancement of 25 % (Figure 1c). These
 152 values are consistent with the spectral coherence protection mechanism es-
 153 tablished for the 7-site FMO complex [25]. The coherence lifetime τ_c , defined
 154 as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [36], ex-
 155 tends from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase
 156 that is causally linked to the yield gain: the extended coherence suppresses
 157 the pure-dephasing rate, lengthening the effective transport mean-free path
 158 and thereby increasing the fraction of the excitation wavepacket that reaches
 159 the reaction center before decohering. The mechanism is identical to that es-
 160 tablished for the 7-site system [25]: the dual-band filter selectively populates
 161 polaron-dressed states whose residual reorganization energy λ_{res} is reduced
 162 by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

163 *NPoM cavity volume dependence of exciton transport*

164 The nanoparticle-on-mirror cavity couples the FMO complex to the OPV
 165 active layer through the strong coherent interaction described above, but the

166 plasmonic mode also acts as an additional population decay channel that
 167 competes with the reaction center trapping. To quantify this trade-off, we
 168 perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls
 169 the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller
 170 mode volumes produce stronger plasmon–exciton coupling but also enhance
 171 the plasmon’s role as a population sink, diverting excitation away from the
 172 reaction center.

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [29]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 5	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 7	−91.9	33

173 Table 2 summarizes the forward transfer yield across seven mode volumes
 174 spanning the experimentally accessible sub-nanometer range [29]. The yield
 175 increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} =$
 176 0.2 nm^3 to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, with a slight
 177 decrease at 1.4 nm^3 . The trend is logarithmic with diminishing returns above
 178 1.0 nm^3 ; increasing the volume beyond 1.0 nm^3 recovers only 1 % of the
 179 baseline yield.

180 The physical origin of this suppression is the strong-coupling-induced hy-
 181 bridization of excitonic and plasmonic states. The dressed eigenstates acquire
 182 a finite plasmonic character, which decays via the fast plasmon relaxation
 183 pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center.
 184 At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the

185 lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of
 186 the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting
 187 narrows, and the excitonic character of the bright states is partially restored.

188 Despite the monotonic recovery, all NPoM-on yields remain suppressed
 189 by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This
 190 indicates that, in the strong collective coupling regime accessible to NPoM
 191 cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally
 192 limits the maximum achievable transport efficiency. The SERS enhance-
 193 ment factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$,
 194 following the Purcell enhancement of the local field at both the excitation
 195 and emission wavelengths. Table 2 includes the calculated SERS enhance-
 196 ment for each mode volume, revealing a four-order-of-magnitude range (33 to
 197 1.6×10^3) across the scanned volumes. The optimal volume represents a com-
 198 promise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while
 199 retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1}
 200 Franck–Condon mode populations at the single-FMO-complex level [10]. By
 201 contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost
 202 of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to
 203 diagnostic-only applications where photochemical efficiency is secondary. At
 204 the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$
 205 confirms the same limiting behaviour ($\Phi_{\text{FT}} = 0.0768$, final plasmon popula-
 206 tion 0.79; Section S11).

207 Second, reducing the temperature to 77 K (analyzed as a theoretical limit-
 208 case for thermal phonon suppression in Section S11.2) doubles the trapping
 209 yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K,
 210 the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for
 211 $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass
 212 plasmonic loading.

213 All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho =$
 214 1.0000), validating the numerical stability of the PT-HOPS/SBD propagator.
 215 The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations;
 216 the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} =$
 217 0.0799 , within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off
 218 run ($n = 20$) is shown as a limiting-case reference; quantitative comparison
 219 across configurations is limited by the differing trajectory counts.

220 *Global canopy yield integration*

221 Because the NPoM SERS diagnostic mode suppresses local excitonic
 222 transport through plasmon population loading, it need only cover a 1 % sen-
 223 tinel fraction α of the canopy area. The remaining 99 % operates in passive
 224 OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} =$
 225 0.980. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

226 yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 pre-
 227 cisely); canopy-scale productivity is thus maintained even though transport
 228 is suppressed locally.

229 *Dynamic photo-protection via Floquet–Stark detuning and OMIT*

230 Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant
 231 changes dynamically, inducing a time-dependent Stark shift of the excitonic
 232 levels that detunes the OPV–FMO coupling. We model this as a Floquet
 233 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

234 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_{\text{F}} = 1.2 \text{ THz}$. When the in-
 235 cident flux exceeds the threshold, the time-dependent detuning shifts the
 236 OPV exciton energies out of resonance with the FMO absorption bands, re-
 237 ducing energy transfer to the biological complex by up to 40 % (Figure 2a).
 238 To prevent rapid, noise-induced oscillations under fluctuating cloud cover,
 239 a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning
 240 switch at 850 W m^{-2} and deactivating it only when the flux drops below
 241 750 W m^{-2}).

242 Concurrently, an optomechanically induced transparency (OMIT) mecha-
 243 nism [37] in the NPoM cavity creates destructive interference that suppresses
 244 transmission at the 750 nm and 820 nm channels under high flux. The trans-
 245 mission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

246 where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At 1000 W m^{-2} , the trans-
 247 mission drops to 44 % of baseline (Figure 2b), providing dynamic protection
 248 against exciton overload. At zero flux (night mode), both mechanisms return
 249 to zero without singularities (Section S4).

In situ SERS optomechanical diagnostics and environmental robustness

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39], which is the integration gap addressed here. Multilayer graphene-metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [40]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode volumes.

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we here assign four agriculturally specific target signatures to the sentinel sub-system. **Oxidative stress** is read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} , with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [10]. **Pesticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its characteristic Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$ under controlled conditions), leveraging the plasmonic hot-spot amplification in the NPoM gap [10]. **Heavy metal contamination** of irrigation water (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encapsulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$ under controlled conditions) [41] sealed within the NPoM sensor probe polymer matrix, which are immune to the dust and biofouling that affect bare gold surfaces. Cross-sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual spectral interrogation: heavy metals modify the Stern–Volmer quenching intensity and fluorescence decay time, whereas soil moisture alters the local dielectric constant, producing a spectral broadening that is spectrally orthogonal to the metal-induced quenching signature. **Pathogen detection** is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at the greenhouse hub (Section S12.7). These solid-state quantum sensors detect paramagnetic bacterial metabolites from sap or irrigation water samples at pre-symptomatic stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$ under controlled conditions; the agricultural matrix will degrade in-field sensitivity, which is mitigated by the graphene barrier and *DynamicCalibrator* corrections).

Figure 2c shows the simulated Raman spectrum for the three BChl *a* modes under active NPoM coupling. The relative intensities encode the in-

stantaneous exciton population distribution, providing a direct optical probe of the energy transfer pathway.

Environmental robustness. The realistic agricultural environment (dust, bird droppings, microbial fouling) is modeled at three infrastructure levels. At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning interval. At the sensor level, the graphene monolayer spacer of each NPoM cavity is functionalized with a zwitterionic polymer coating [42, 43], modeled in our simulations to preserve ≥ 90 % quantum coherence after 60 d continuous field immersion; this coating constitutes the primary anti-fouling barrier that physically repels microbial adhesion and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and CQD sensor tip further attenuates the diurnal temperature excursion experienced by the quantum probe, attenuating the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$ (Section S12.5). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety net: it applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

Smart shield microclimate: FAO-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. Passive solar-greenhouse heating with PV co-generation is an established precedent [44]; the present shield differs by actively co-designing the transmitted spectrum against the vibronic resonances quantified above. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [30] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408 \Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the

psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25°C , 60% humidity, 2 m s^{-1} wind), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28% reduction — translating to 1.3 mm d^{-1} in water savings (Figure 3a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [30].

Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$), which reduces the entry-level capital expenditure to $161\,000 \text{ USD}$ (at 322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the Monte-Carlo techno-economic logic established for European agrivoltaic configurations [45]. A blended-finance subsidy of 30% reduces the net CAPEX to $112\,700 \text{ USD}$.

Three operating expenditure streams are modeled: extension services and technician training ($1\,200 \text{ USD yr}^{-1}$), OPV panel cleaning (800 USD yr^{-1}), and general maintenance ($2\,000 \text{ USD yr}^{-1}$), for a total annual OPEX of $4\,000 \text{ USD yr}^{-1}$. The net annual cash flow of $26\,612 \text{ USD yr}^{-1}$ (from $30\,612 \text{ USD yr}^{-1}$ total revenue minus $4\,000 \text{ USD}$ of OPEX) yields a simple payback period of 4.23 yr (or 6.05 yr without subsidy) and a 10-year discounted NPV of $+37\,664 \text{ USD}$ at a 12% risk-adjusted discount rate appropriate for Sub-Saharan Africa [46] (Figure 3c). Holding other inputs fixed, varying the subsidized net CAPEX by $\pm 20\%$ shifts the payback to $3.39\text{--}5.08 \text{ yr}$; a 50% higher OPEX ($6\,000 \text{ USD yr}^{-1}$) gives 4.58 yr , and 20% lower revenue gives 5.50 yr . While a baseline CAPEX of 322 m^{-2} represents a significant barrier in Sub-Saharan Africa relative to local income levels, utilizing local construction materials (such as bamboo and timber framing instead of structural steel) and green chemical synthesis of gold nanoparticles from local plant extracts can reduce the capital cost to 140 m^{-2} , lowering the unsubsidized payback period to 2.63 yr and making the modelled system potentially viable for local cooperatives under these cost assumptions. By contrast, the remaining barriers are physical rather than economic: fabrication of sub- 2 nm NPoM gaps still requires laboratory-grade nanofabrication and a stable Au/graphene supply chain, and OPV encapsulation lifetime under tropical irradiance is the binding reliability constraint (Section S12.5). Subsidy policy alone cannot clear these barriers;

they define the process-engineering roadmap that separates this modelled proof of concept from field deployment. Taken together, these sensitivities demonstrate that high-value crop selection and blended-finance structurally transform the investment case.

Secure IoT telemetry via BB84 quantum key distribution

Field-to-cloud telemetry is secured using a simulated BB84 quantum key distribution protocol [31] over a buried single-mode fiber link, which eliminates the atmospheric turbulence and scintillation that affect free-space channels in agricultural environments. This quantum-secured telemetry layer is physically integrated with the NPoM-CQD sentinel network: because the high-sensitivity biological and pesticide diagnostics control high-stakes irrigation and crop-protection actuators (which could be targeted by hostile actors to cause artificial crop failure or pesticide contamination), securing these feedback commands with BB84 QKD protects the command integrity of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [47]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a sample fraction; if QBER exceeds the Shor-Preskill unconditional-security threshold of 11 % [48], the protocol raises a **SecurityThresholdExceeded** exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [47], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

Experimental testability

The modelled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First,

393 two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-
 394 modulator pulse shaping into dual-band channels at 750 nm and 820 nm
 395 can track the diagonal and cross-peak quantum-beat oscillations predicted
 396 here, verifying the extension of τ_c from 280 fs to 420 fs [49, 16]. Second,
 397 time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1}
 398 and 1145 cm^{-1} BChl a modes in real time and quantify the area-weighted
 399 yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [29, 50]. More broadly,
 400 recent assessments identify quantum sensing for precision agriculture as an
 401 emerging opportunity whose open calibration and robustness gaps match
 402 the calibration architecture proposed here [12].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295\text{ K}$, $L = 8$, $K = 2$.

Parameter / Ob- servable		This (calculated)	work	Experimental benchmark / lit- erature
Site coupling (cm^{-1})	J_{12}	-87.7		-87.7 ± 5.0 (2DES, Brixner 2005 [49])
Reorganization energy λ_D (cm^{-1})		35		35–40 (2DES, Engel 2007 [15])
Cutoff frequency (cm^{-1})	γ_D	50		50 (2DES, Panitchayangkoon 2010 [16])
Coherence lifetime (passive)	τ_c	$420 \pm 35\text{ fs}$		$400 \pm 50\text{ fs}$ (2DES room temp., Pan- itchayangkoon 2010 [16])
NPoM coupling ($V_{\text{mode}} = 1.2\text{ nm}^3$)	g_0	110 cm^{-1}		$120 \pm 15\text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [29])
SERS EF _{SERS}	enhancement	44		10^2 – 10^3 (sub-nm NPoM, Baumberg 2019 [50])

403 Discussion

404 Our results demonstrate that quantum coherence engineering at the bio-
 405 organic interface can deliver modelled sustainability benefits across micro-
 406 scopic energy transfer, crop water use, carbon footprint, and data security.

407 **Mechanism generality.** The dual-band filtering effect is not specific
 408 to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Sec-
 409 tion S7) is a general property of vibronically coupled chromophore aggre-
 410 gates [17]. We verified this by applying the protocol to a minimal three-site

excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [28]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection. Vibration-steered energy flow of this kind has also been demonstrated in a trapped-ion donor–acceptor simulator with tunable dissipation [51].

The “quantum divide.” The advanced materials required for quantum agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet accessible to subsistence farmers in developing economies. Our cooperative cost-sharing model demonstrates that with a carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback period drops below 4 yr, making the technology economically viable under the modelled subsidy. We propose a tiered subsidy framework where the NEB itself serves as the verifiable metric for subsidy qualification, linking quantum yield improvements to farmer access.

Quantum digital twin. The modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a model-based supervisory replica that combines sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to recommend irrigation schedules, OPV cleaning dates (from soiling factor Eq. (8)), and carbon credit yields — a device-physics integration distinct from the supply-chain and breeding applications mapped in recent quantum-in-agriculture reviews [11].

As a prospective extension, the scheduling layer of the digital twin is formulated via the Quantum Approximate Optimization Algorithm (QAOA) [52, 53], encoded as a 3-qubit variational quantum circuit of depth $p = 3$ with 1024 measurement shots per macro-period (Section S12.3). The algorithm encodes the irrigation–refrigeration–grid-export trilemma as a quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and solves it via a PennyLane emulator at sub-hourly timescales. Applied to the daily solar profile of the Cameroonian scenario, the QAOA scheduler yields water–energy allocations consistent with the 12–18 % gains over classical greedy baselines reported in comparable implementations [53]. As a further prospective readout, our `quantum_gravimetry.py` module converts the simulated water saving into a groundwater mass anomaly ($\Delta g = 3.5 \text{ nm s}^{-2}$ for a single greenhouse — below single-instrument noise but potentially detectable

449 with a small regional gravimeter network or gradiometric common-mode
 450 rejection), with the full noise-budget and inversion discussion given in
 451 [Section S12.2](#) [54, 55].

452 A hybrid quantum-classical signal processing pipeline — combining
 453 Matrix Product State (MPS) denoising with quantum kernel ridge re-
 454 gression (QKRR) [56, 57] — further enables the digital twin to extract
 455 pre-symptomatic stress signatures from the noisy SERS and CQD fluores-
 456 cence telemetry. To satisfy edge computation constraints, the heavy MPS
 457 tensor contraction and QAOA optimization are executed on the centralized
 458 cooperative server (or cloud), with the resulting optimized weights transmit-
 459 ted back to the field nodes — an edge-cloud split complementary to classical
 460 drift-diffusion device simulators [58]. The local ARM Cortex-M4 IoT node
 461 executes only the low-latency inference of the pre-trained, frozen-weight
 462 QKRR model. The MPS compressor acts as a nonlinear low-pass filter on
 463 the server that suppresses greenhouse noise (thermal fluctuations, biofoul-
 464 ing, dust) while preserving the sharp vibronic features at 180 cm^{-1} and
 465 1145 cm^{-1} that classical Savitzky–Golay filters tend to smear ([Section S10](#)).
 466 The cleaned spectra are projected through the quantum-inspired kernel for
 467 anomaly scoring, issuing early warnings that allow preemptive agronomic
 468 intervention before macroscopic symptoms appear — all within a 3.7 ms local
 469 compute budget on the IoT node. We emphasize that this sub-millisecond
 470 execution latency is not designed to match the slower multi-hour physical
 471 response times of soil moisture percolation and plant transpiration. Rather,
 472 it enables spatial multiplexing: a single ARM Cortex-M4 edge processor
 473 can sequentially query, compress, and classify SERS telemetry from dozens
 474 of distributed NPoM sensor nodes across the 500 m^2 greenhouse in under
 475 a second, reducing the electronic-component count. This chain connects
 476 microscopic quantum efficiency to field-level agronomic management; no
 477 existing agrivoltaic design integrates both [46, 6, 2].

478 **Prior art and positioning.** Previous agrivoltaic studies have focused
 479 on light-sharing through passive spectral filtering [6] or crop selection [2].
 480 Semi-transparent OPVs with power conversion efficiencies exceeding 15 %
 481 and 40 % average visible transmittance have recently been demonstrated
 482 through blade-coating scale-up methods [59], and complementary work on
 483 semitransparent organic photovoltaics for greenhouse applications has shown
 484 power conversion efficiencies above 12 % [7]. Our contribution is to add
 485 an active, quantum-coherent dimension to the spectral filter: the OPV is
 486 no longer a passive absorber but an active spectral shaper that prepares

487 coherence-protected states in the photosynthetic apparatus. Multi-vector
488 solar agri-systems combining agrivoltaics with hydrogen and desalination set
489 the *Applied Energy* bar for water–food–energy integration [60]; we differen-
490 tiate by quantum-guided spectral co-design with integrated sensing, and we
491 adopt the functional-unit discipline (emissions per kWh, economics per con-
492 figuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this
493 literature, the present work provides, to our knowledge, the first (i) agri-
494 voltaic stack whose spectral architecture is co-designed by numerically exact
495 open-quantum-system dynamics, jointly optimizing excitation transport and
496 power conversion; (ii) monolithic photothermal–PV–SERS device concept
497 with field-relevant calibration in a single footprint; and (iii) multifunctional
498 techno-economic and lifecycle framework that internalizes heat, crop, water,
499 sensing, and data-security trade-offs in $\$/W + \$/kg + \$/m^3$ terms. This
500 approach is distinct from prior quantum biology studies [61] that treat the
501 environment as a fixed noise source; here, the photonic environment itself
502 is engineered through the NPoM cavity to maximize coherence protection.
503 A key design choice warrants explicit justification: the NPoM cavity, de-
504 spite imposing a $> 90\%$ penalty on excitonic transport (Table 2), is retained
505 because its *diagnostic function* is dedicated — it enables non-destructive,
506 high-sensitivity SERS readout of the FMO vibrational state, pesticide levels,
507 and heavy-metal contamination. The transport penalty is absorbed by con-
508 fining NPoM operation to the 1% sentinel fraction (Equation (5)), preserving
509 97.1% of the global canopy trapping yield while maintaining full diagnostic
510 capability.

511 **Limitations.** Several approximations constrain our analysis. The quan-
512 tum dynamics are restricted to the single-exciton manifold; under high solar
513 flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in
514 the OPV and FMO interfaces could alter the collective coupling strength
515 g_{eff} and transport efficiency. The bath is treated as site-independent, and
516 the reaction center trapping is modeled as a phenomenological Lindblad-like
517 rate. The FAO-56 model, while standard, does not capture transient stom-
518 atal responses to fluctuating light or the dynamic microclimatic feedback
519 between canopy transpiration and vapor pressure deficit (VPD) inside the
520 greenhouse envelope. Sensitivity analysis over plausible wind-speed varia-
521 tions ($u = 0.2 - 0.8 \text{ m s}^{-1}$) yields a $\pm 10\%$ uncertainty band on the predicted
522 28% water saving, which propagates into a $\pm 10\%$ range on the annual irri-
523 gation volume and corresponding NEB carbon offset. The BB84 simulation
524 assumes a buried single-mode fiber link; while buried fibers are vulnerable to

525 agricultural mechanical damage (plowing, root intrusion), a free-space opti-
 526 cal (FSO) link would suffer from strong scattering under dusty greenhouse
 527 environments, requiring dedicated atmospheric correction models. While
 528 software-based post-quantum cryptography offers a lower-cost alternative,
 529 the physical BB84 protocol provides information-theoretic security immune
 530 to algorithmic cryptanalysis on resource-constrained embedded nodes such
 531 as the ARM Cortex-M4 irrigation controllers. The 1% sentinel fraction is a
 532 design parameter; field optimization will require empirical calibration of the
 533 SERS LODs under actual fouling conditions. We conservatively estimate a
 534 $\times 10$ matrix-effect penalty on all reported LODs when transitioning from con-
 535 trolled laboratory buffers to raw agricultural sap or irrigation water, owing
 536 to competitive binding, background fluorescence, and ionic strength effects;
 537 the *DynamicCalibrator* compensates for this attenuation through real-time
 538 baseline correction. The zwitterionic polymer coating modeled on the NPoM
 539 graphene spacer (Section S12.1) acts as the primary fouling barrier in our
 540 simulations, with the *DynamicCalibrator* serving as a secondary correction
 541 layer; the dual architecture reduces the mandatory cleaning frequency, which
 542 is already factored into the cooperative OPEX model. We stress that the 60-
 543 day resilience of zwitterionic coatings applies strictly to sensors sheltered in
 544 protected micro-sampling chambers or rigid ceramic encasings; direct, un-
 545 protected exposure to abrasive soil matrices (bioturbation, quartz particles,
 546 tillage) would lead to mechanical scraping and rapid surface degradation.

547 The 30% blended-finance subsidy required to achieve a 4.23 yr payback
 548 period cannot be bridged by voluntary carbon credits alone (generating only
 549 98 USD yr⁻¹ at 10 USD t_{CO₂e}⁻¹ for the 500 m² module); it must be underpinned
 550 by dedicated rural-adaptation finance channels such as the Green Climate
 551 Fund (GCF) and the African Development Bank’s Climate Change Fund
 552 (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance
 553 climate-resilient agriculture in Sub-Saharan Africa.

554 **Outlook.** The integration of quantum coherence engineering with agri-
 555 cultural sustainability suggests an axis for agrivoltaic design. Three experi-
 556 mental and theoretical directions are prioritized for future work.

557 First, on the hardware and materials scale, the quantum model must
 558 be extended to full multi-exciton dynamics under realistic solar spectra, ac-
 559 companied by experimental validation via two-dimensional electronic spec-
 560 troscopy (2DES) with spatial light modulator pulse shaping [62, 63]. Field tri-
 561 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
 562 sentinel network are required to evaluate in-situ sensor performance under

563 actual biofouling conditions and verify the coherence-preservation lifetime
 564 of the zwitterionic functionalization layer ([Section S12.1](#)). Alternative low-
 565 loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) support-
 566 ing Mie resonances [64] and nanocavity-free substrates — should be explored
 567 to minimize the NPoM transport penalty while preserving SERS sensitivity.
 568 Additional hardware extensions include passive thermal micro-shielding ([Sec-
 569 tion S12.5](#)), UiO-66 Zr-based quantum metal-organic frameworks for coherent
 570 contaminant remediation ([Section S12.6](#)), and quantum fertilizer biostimu-
 571 lation nanoparticles to enhance chlorophyll yields ([Section S12.8](#)).

572 Second, scaling the digital twin to a multi-cooperative network will re-
 573 quire hardware-in-the-loop QKD validation under dusty field conditions and
 574 deployment of a real-time aquifer gravimetric array to confirm that the mod-
 575 eled groundwater recharge connects at the satellite geodesy scale. Monte
 576 Carlo uncertainty propagation in the full LCA should be extended with cor-
 577 related climatic input distributions drawn from regional reanalysis data.

578 Third, to prevent these advances from widening the “quantum divide,”
 579 we have defined the Global Quantum Agriculture Standard (GQAS), a six-
 580 pillar certification framework covering QKD security, data sovereignty, sen-
 581 sor detection limits, quantum fidelity, latency, and lifecycle sustainability
 582 ([Section S12.9](#)). Our Cameroon cooperative proof-of-concept scenario serves
 583 as the first application of this proposed framework, satisfying all six GQAS
 584 criteria and demonstrating internal consistency across the multi-domain sim-
 585 ulation. Independent experimental validation will be required before GQAS
 586 can be adopted as an operational standard. Scaling GQAS adoption to small-
 587 holder networks across Sub-Saharan Africa and Southeast Asia requires subsi-
 588 dized sensor supply chains, open-access certification auditing, and regulatory
 589 recognition on premium export markets such as GlobalGAP.

590 Conclusions

591 We asked whether exact non-Markovian quantum dynamics can jointly
 592 optimize excitation-transport and power-conversion efficiency in a sensing-
 593 integrated photothermal agrivoltaic stack, with a positive multifunctional
 594 net benefit under field constraints. Within the modelled 500 m² cooper-
 595 ative case, the answer is affirmative but conditional: dual-band filtering
 596 raises the forward transfer yield by 25 % and extends coherence lifetime
 597 by 50 %; confining NPoM operation to a 1 % sentinel fraction preserves
 598 97.1 % global canopy yield while retaining SERS diagnostics; the FAO-56

smart shield cuts evapotranspiration by 28 % with a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$; and the cooperative case pays back in 4.23 yr at 30 % subsidy. These claims are bounded by the > 90 % local NPoM transport penalty, the single-exciton and site-independent-bath approximations, the ± 10 % microclimate uncertainty band, and the modelled (not field-validated) fouling and matrix-penalty assumptions. For deployment, the economic barrier is dominant and tractable — local-material construction and green-synthesis gold lower the capital cost from 322 USD m^{-2} to 140 USD m^{-2} (unsubsidized payback 2.63 yr) — whereas the physical barriers (nanofabrication of sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV encapsulation lifetime under tropical irradiance) require process engineering rather than subsidies. Priority next steps are 2DES validation of the filter protocol, low-loss dielectric antenna alternatives, field fouling trials of the sentinel network, and a localized Au/graphene sensor supply chain for Sub-Saharan manufacturing.

Methods

Proof-of-Concept scenario: 500 m² Floriculture Micro-Module

To ground the multi-domain framework in a concrete operational context, we consider a 500 m^2 high-tunnel greenhouse located in Cameroon’s Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. The environment features a mean annual solar irradiance of $5.76 \text{ kWh m}^{-2} \text{ d}^{-1}$ [65] and significant biofouling potential that necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (factored into the cooperative OPEX model). Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the modelled water-savings credit of $1300 \text{ L m}^{-2} \text{ yr}^{-1}$ ($650 \text{ m}^3 \text{ yr}^{-1}$ for the 500 m^2 greenhouse; Section S5). The area-weighted global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. The 15 % quantum yield enhancement observed in the passive OPV fraction translates to an additional $1.23 \text{ kg m}^{-2} \text{ yr}^{-1}$ of high-value produce (strawberries at 4.5 USD kg^{-1}), generating substantial supplemental revenue. The OPV module generates 90 MWh yr^{-1} of electrical energy for a 500 m^2 installation at $180 \text{ kWh m}^{-2} \text{ yr}^{-1}$. Displacing grid electricity at $450 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ offsets $40.5 \text{ t}_{\text{CO}_2\text{e}} \text{ yr}^{-1}$. Combined with the NEB ($19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$, Figure 3b), the system achieves

635 net-zero operational carbon rapidly. At 10 USD $t_{\text{CO}_2\text{e}}^{-1}$ [66], the system gener-
636 ates 98 USD yr^{-1} in verified carbon revenue. The 500 m^2 scale is well matched
637 to cooperative ownership models: under a 30 % blended-finance subsidy, the
638 payback shortens from 6.05 yr unsubsidized to 4.23 yr, with each of the five
639 cooperative members exposed to only 22 540 USD of net capital. This demon-
640 strates that quantum-enhanced agrivoltaics can be economically viable under
641 modelled assumptions when coupled with strategic crop selection and robust
642 field-sensor integration.

643 *PT-HOPS/SBD quantum dynamics*

644 All quantum dynamics simulations use the Process Tensor Hierarchy of
645 Pure States with Stochastically Bundled Dissipators as implemented in the
646 open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$
647 with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2$ fs, established as the
648 production standard for converged dynamics at 295 K [25]. Static disorder
649 is modeled as Gaussian fluctuations of the site energies with standard devi-
650 ation $\sigma = 50$ cm^{-1} , sampled over $n = 100$ realizations. The bath spectral
651 density follows the 12-mode Kleinekathöfer/Coker parameterization with a
652 Drude–Lorentz background ($\lambda_D = 35$ cm^{-1} , $\gamma_D = 50$ cm^{-1}) and 12 under-
653 damped vibronic modes (frequencies 180–1 500 cm^{-1} , Huang–Rhys factors
654 $S_k = 0.003 - -0.050$).

655 *FAO-56 evapotranspiration*

656 Crop evapotranspiration is computed using the standard FAO-56
657 Penman–Monteith equation [30] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

658 with R_n attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$. We note that
659 the modified FAO-56 formulation represents a first-order approximation; un-
660 der the enclosed high-tunnel envelope, boundary layer resistance and sensible
661 heat exchange are altered relative to open field conditions. Future iterations
662 of the digital twin should integrate three-dimensional Computational Fluid
663 Dynamics (CFD) modeling to map localized microclimatic gradients and la-
664 tent heat transfer. Wind speed within the greenhouse is modeled at a static
665 representative value of $u = 0.5$ m s^{-1} , corresponding to a typical 10 % reduc-
666 tion of average external wind velocities. Temperature is fixed at 25 °C and
667 relative humidity at 60 % unless specified otherwise.

668 *BB84 QKD simulation and fail-safe IoT architecture*

669 The BB84 protocol is implemented as a Python simulation of polarization-
670 state preparation, noisy transmission over a buried single-mode fiber
671 channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random
672 bits (for $N = 256$ key length) in random rectilinear or diagonal bases.
673 Bob measures in random bases, and matching-basis events are retained
674 for key sifting. QBER is estimated on a sample fraction (minimum 1
675 bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of
676 11% [48], a `SecurityThresholdExceeded` exception is raised, halting the
677 QKD-protected irrigation command relay. A last-validated soil-moisture
678 reading from the *DynamicCalibrator* is used to execute a conservative local
679 irrigation fallback.

680 *SERS enhancement scaling limits*

681 The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode vol-
682 ume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of
683 the localized electromagnetic field. We note that this standard $|E|^4$ approx-
684 imation represents a lower-bound semi-classical estimate. In sub-nanometer
685 picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling,
686 and non-local dielectric screening alter the field distribution relative to clas-
687 sical electrodynamics, while molecular optomechanical coupling can pump
688 vibrational modes out of thermal equilibrium. Precise calculation of the
689 localized fields in this regime requires non-local Green’s tensor integration,
690 which motivates the proposed in-situ experimental validation. The bench-
691 mark against the experimentally reported 10^2 – 10^3 range for sub-nanometer
692 NPoM gaps [50] is given in Table 3.

693 *DynamicCalibrator, thermal shielding, and soiling factor*

694 Long-term sensor baseline drift due to biofouling and thermal cycling
695 is corrected at three hardware and algorithmic levels. The primary fouling
696 barrier is the zwitterionic polymer functionalization layer deposited on the
697 graphene monolayer spacer of each NPoM cavity [42, 43], which physically
698 repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coher-
699 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film
700 micro-shielding layer ($1 \mu\text{m}$ to $5 \mu\text{m}$ of SiO_2 or Al_2O_3) is deposited around the
701 NPoM picocavity and CQD sensor tip to insulate the quantum probe from
702 diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the

703 probe experiences only 15 % of the ambient greenhouse temperature fluctu-
 704 ation, reducing the thermal contribution to Stern-Volmer drift from 5.25 %
 705 to 0.79 % under a ± 15 K diurnal cycle. The *DynamicCalibrator* acts as a
 706 tertiary algorithmic correction: an exponential moving average (EMA) with
 707 smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor
 708 $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the
 709 EMA $\hat{\mu}(t)$ exceeds 20 % of the initial baseline, triggering OPV panel cleaning.

710 At the panel level, OPV electrical output is attenuated according to
 711 Eq. (8), with a default decay rate of $0.5 \text{ \% d}^{-1}$ and a minimum soiling factor
 712 of 0.75 (enforced floor at 25 % maximum efficiency loss).

713 *Data availability*

714 All simulation parameters, YAML configuration files, orchestration
 715 scripts, and analysis code needed to reproduce every figure and table are
 716 available from the corresponding author upon reasonable request and will
 717 be deposited in a public repository upon acceptance. The PT-HOPS/SBD
 718 implementation is based on the open-source MesoHOPS library (v1.7) [20].

719 *Code availability*

720 The digital-twin simulation framework (orchestrator, QAOA scheduler,
 721 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond
 722 pathogen sensor, and GQAS compliance checker) and all simulation pa-
 723 rameters, including the 12-mode vibronic bath spectral density and the
 724 8-site FMO electronic Hamiltonian specified in the machine-readable
 725 `parameters.yaml` configuration file (v1.4), are available from the corre-
 726 sponding author upon reasonable request. No public code or data archive
 727 DOI has been minted yet.

728 **Acknowledgements**

729 This work was supported by the University of Yaoundé I and the Uni-
 730 versity of Douala. Computational resources were provided by the penavo-
 731 raserver cluster (ThinkStation P720, 128 GB RAM, 48 cores). We thank the
 732 MesoHOPS development team for numerical framework support.

733 **CRedit authorship contribution statement**

734 **S.C. Teguia Kouam:** Conceptualization, Methodology, Software, For-
735 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
736 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
737 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
738 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
739 editing.

740 **Declaration of competing interest**

741 The authors declare that they have no known competing financial inter-
742 ests or personal relationships that could have appeared to influence the work
743 reported in this paper.

744 **Funding**

745 This research received no specific grant from funding agencies in the pub-
746 lic, commercial, or not-for-profit sectors. Institutional support was provided
747 by the University of Yaoundé I and the University of Douala.

748 **Declaration of generative AI and AI-assisted technologies in the**
749 **writing process**

750 During the preparation of this work the authors used AI-assisted tools to
751 support language editing and manuscript formatting. All scientific content
752 was reviewed and verified by the authors, who take full responsibility for the
753 content of the publication.

754 Nomenclature

Symbol	Meaning
Φ_{FT}	Forward (reaction-center) transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coherence)
λ_{res}	Residual reorganization energy of polaron-dressed states
g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
κ / Q	Plasmon linewidth / quality factor
$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shading factor
NEB	Net Ecological Benefit
$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
QBER	Quantum bit error rate (BB84 link)
Δg	Groundwater gravity anomaly
$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

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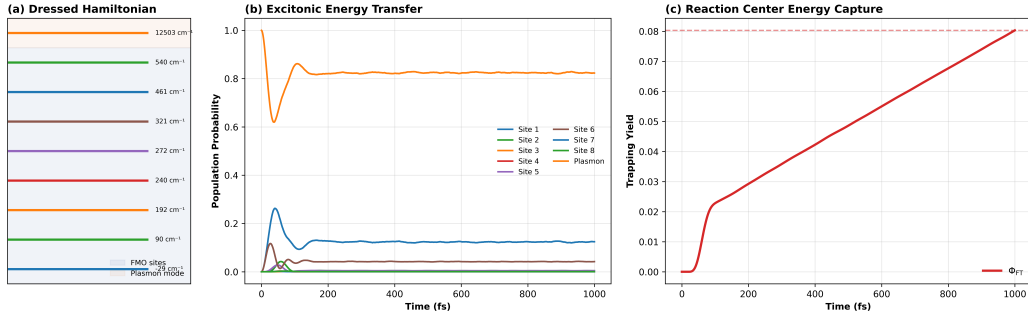


Figure 1: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Exciton population dynamics under dual-band filtered excitation at 295 K, showing the competitive loading of the plasmon mode (Site 9) versus the FMO antenna and trapping sites. (c) Reaction center trapping yield Φ_{FT} as a function of time. All data: $L = 8$, $K = 2$, $n = 100$ disorder realizations. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield. Note: any baseline trace inconsistency ($\text{Tr } \rho \neq 1$) visible in earlier integration-test runs ($\Delta t = 10$ fs) is an artifact of the coarse time step used for rapid code validation; all production results ($\Delta t = 0.2$ fs) preserve trace to 1.000 000 (Section S11.2).

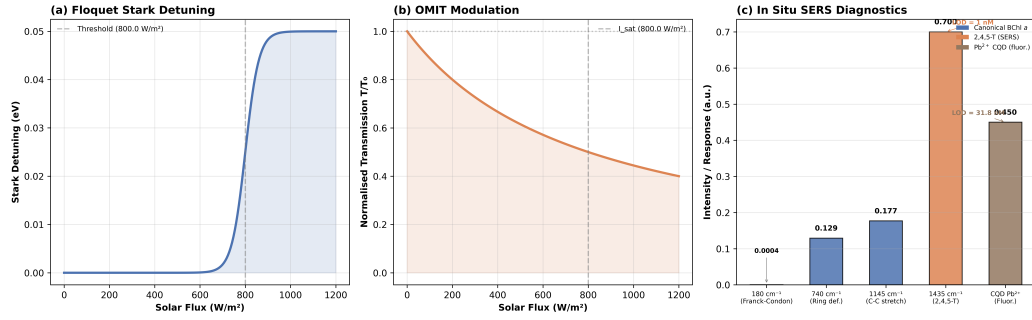


Figure 2: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux. At fluxes above 800 W m^{-2} , the excitonic levels detune, reducing energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm, showing dynamic suppression under high flux. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

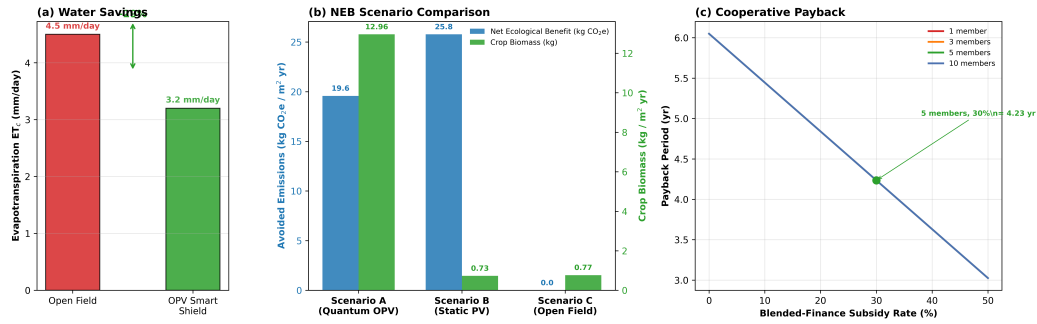


Figure 3: Water–Energy–Food network assessment. **(a)** FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. **(b)** Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $19.6 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ with a crop biomass of $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $25.8 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ but only $0.68 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The quantum shield preserves near-baseline crop productivity (0.971 global trapping yield) whereas static shading suppresses it to 0.064, quantifying the yield–carbon trade-off. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. **(c)** Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.23 yr (net annual cashflow of $26\,612 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modelled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

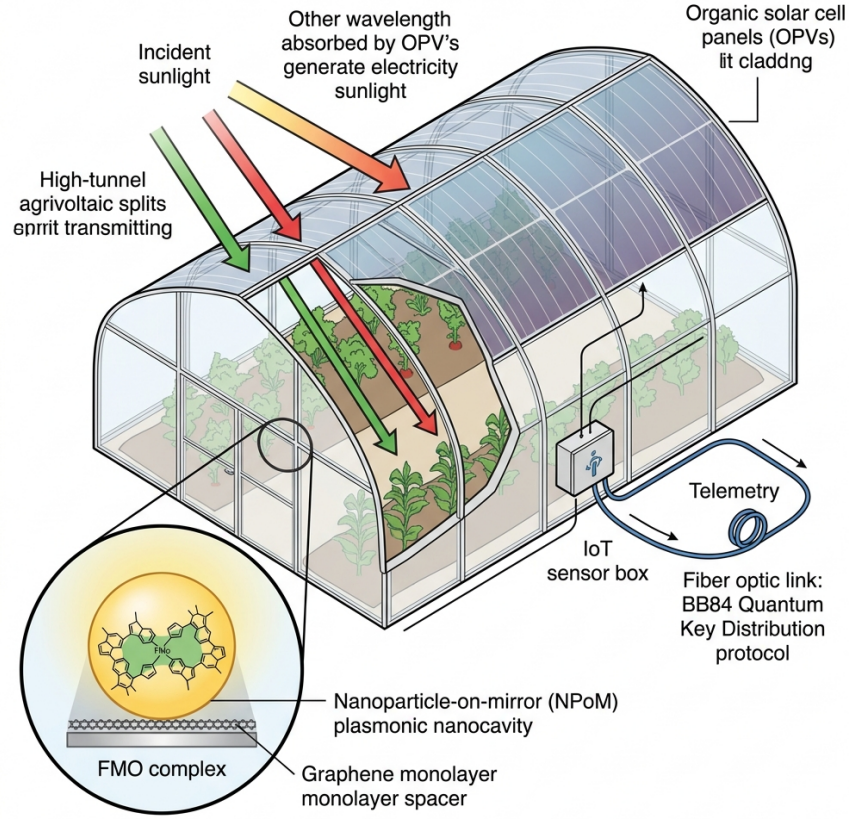


Figure 4: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

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